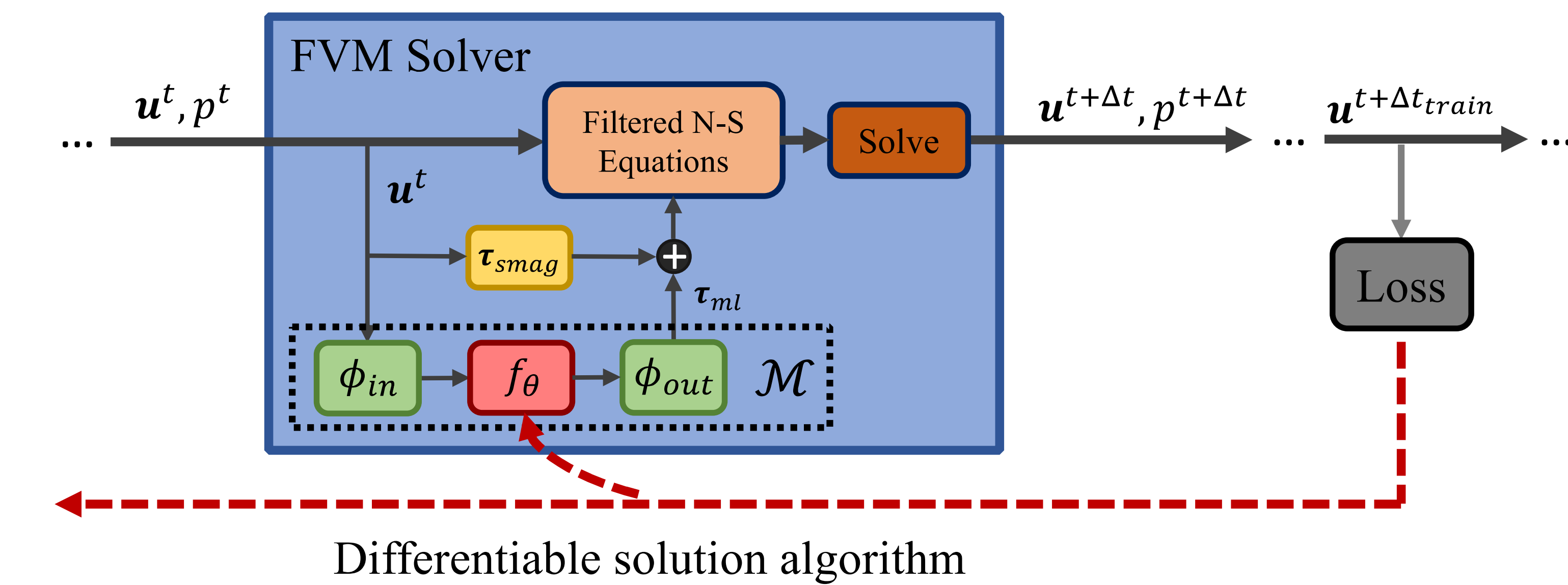


- Aim** – To improve the accuracy of sub-grid scale (SGS) turbulence closure models for Large Eddy Simulations (LES)
- Method** - end-to-end differentiable solver used in combination with physics-inspired deep learning architectures.
- Outcome** –
 - Generalization to a variety of flow configurations, including higher Reynolds numbers and different forcing conditions.
 - Superior performance to offline, *a-priori* learning without necessity of fully resolved simulations



Methodology

- LES equations need closure models to account for unresolved length scales due to G_Δ (subsampling/filtering).

$$\bar{\mathbf{u}}(x, t) = G_\Delta \star \mathbf{u}(x, t)$$

$$\frac{\partial \bar{\mathbf{u}}}{\partial t} + \nabla \cdot (\bar{\mathbf{u}} \otimes \bar{\mathbf{u}}) = \frac{1}{Re} \nabla^2 \bar{\mathbf{u}} - \frac{1}{\rho} \nabla \bar{p} + \bar{\mathbf{f}} + \nabla \cdot \boldsymbol{\tau}$$

- Data-driven methods can be used to learn such closures – either using DNS data (*a-priori*) or using differentiable closure modeling (*a-posteriori*).
- τ_{ij} represents the effects of the unresolved velocity components on the resolved field and is called the Sub-Grid Scale (SGS) stress.

$$\tau_{ij} = \bar{u}_i \bar{u}_j - \bar{u}_i \bar{u}_j \quad \nu_t = (C_s \Delta)^2 |\bar{\mathbf{S}}|$$

$$\hat{\boldsymbol{\tau}} = \boldsymbol{\tau}_{smag} + \boldsymbol{\tau}_{ml} \quad \boldsymbol{\tau}_{smag} = -2\nu_t \bar{\mathbf{S}},$$

- A dimensionless empirical coefficient C_s , a characteristic length scale Δ and a characteristic velocity $\Delta|\mathbf{S}|$, where $|\mathbf{S}| = (2S_{ij}S_{ij})^{1/2}$, define the benchmark Smagorinsky model.
- Using inputs as strain and rotation rate tensors $\mathbf{S}$ and $\mathbf{R}$, we use two approaches :

- Using Tensor Basis – $\mathbf{T}^{(0)} = \mathbf{I}$, $\mathbf{T}^{(1)} = \mathbf{S}$, $\mathbf{T}^{(2)} = \mathbf{SR} - \mathbf{RS}$.

$$\boldsymbol{\tau}_{ml} = \sum_n \alpha^{(n)} \mathbf{T}^{(n)}$$

- Directly Learning the isotropic, deviatoric, and asymmetric components –

$$\boldsymbol{\tau}_{ml} = \alpha \mathbf{I} + \mathbf{D} + \mathbf{A}$$

TABLE I. Model names

Name	ϕ_{out}
Linear (LIN)	$\sum_{n=0}^1 \alpha^{(n)} \mathbf{T}^{(n)}$
Non-linear (NL)	$\sum_{n=0}^2 \alpha^{(n)} \mathbf{T}^{(n)}$
Non-linear asymmetric (NLA)	$\sum_{n=0}^3 \alpha^{(n)} \mathbf{T}^{(n)}$
Model-free (MF)	$\alpha \mathbf{I} + \mathbf{D} + \mathbf{A}$
Model-free symmetric (MFS)	$\alpha \mathbf{I} + \mathbf{D}$

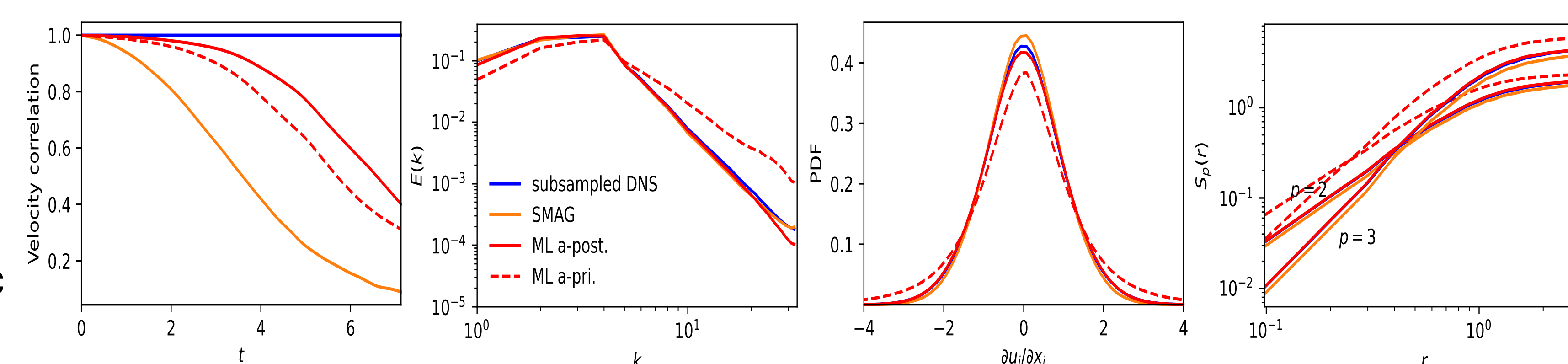
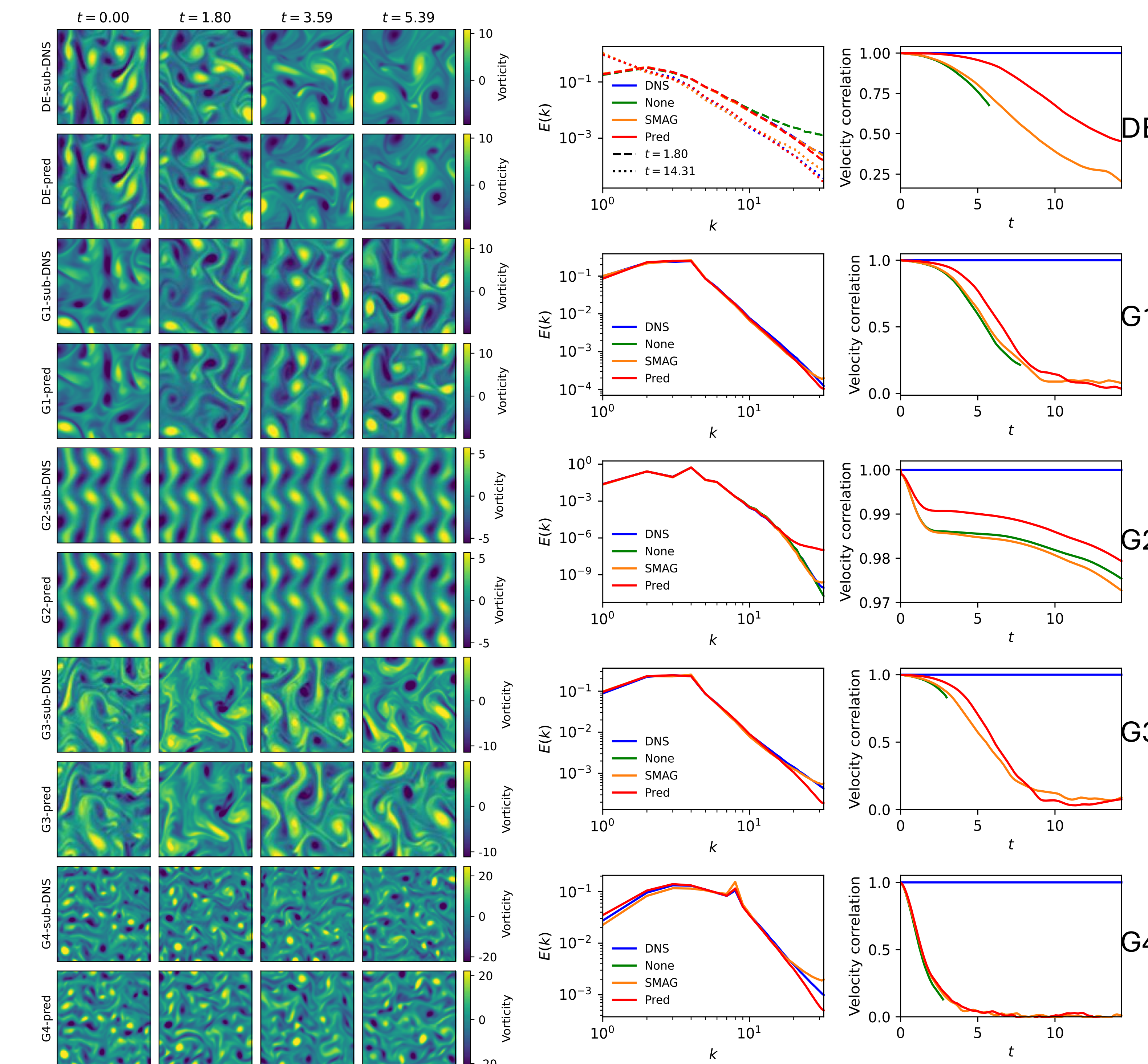
TABLE II. Dataset parameters

Dataset	Re	A	k	r	$\mathbf{f}$	direction
DE	1000	0	0	0	$\hat{\mathbf{e}}_x$	
G1	1000	1	4	0.1	$\hat{\mathbf{e}}_x$	Test
G2	30	1	4	0.1	$\hat{\mathbf{e}}_x$	
G3	10^5	1	4	0.1	$\hat{\mathbf{e}}_x$	
G4	8000	2	8	0.1	$\hat{\mathbf{e}}_y$	

Training Details

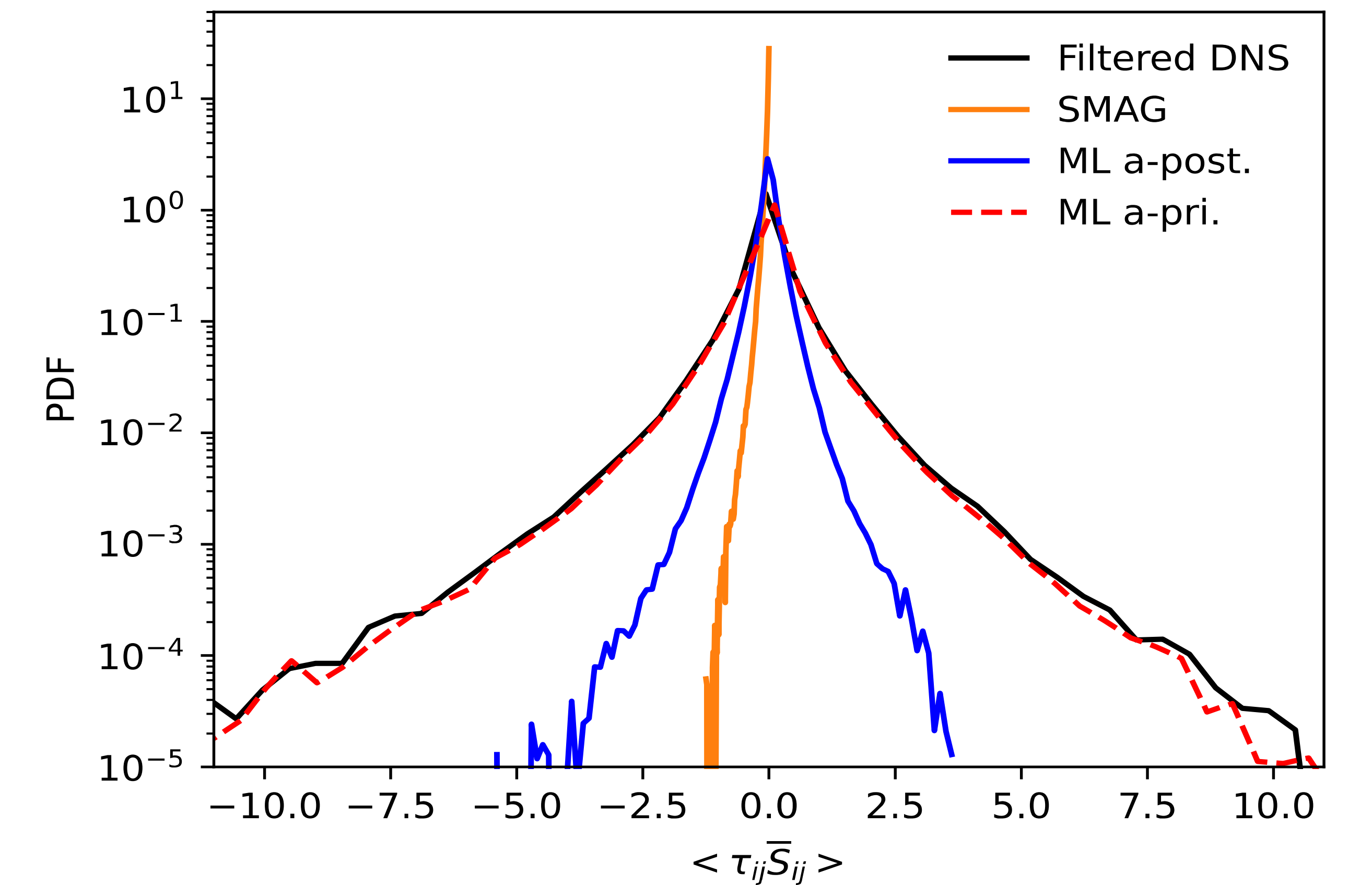
- Architectures used are Convolutional Neural Networks (CNN), Fourier Neural Operators (FNO), Multi-layer Perceptrons (MLP), and a hybrid CNN+FNO.
- Mean-squared-error (MSE) loss is minimized to obtain the optimal parameters of the model.
- 2 hours training time on 4 A100 GPUs and 300 epochs.

Results



A-posteriori model performance is superior to the *a-priori* for correlation. The *a-posteriori* model displays no noticeable disagreement with the DNS statistics, in contrast to the *a-priori* model. SMAG captures statistics but has poor instantaneous correlation.

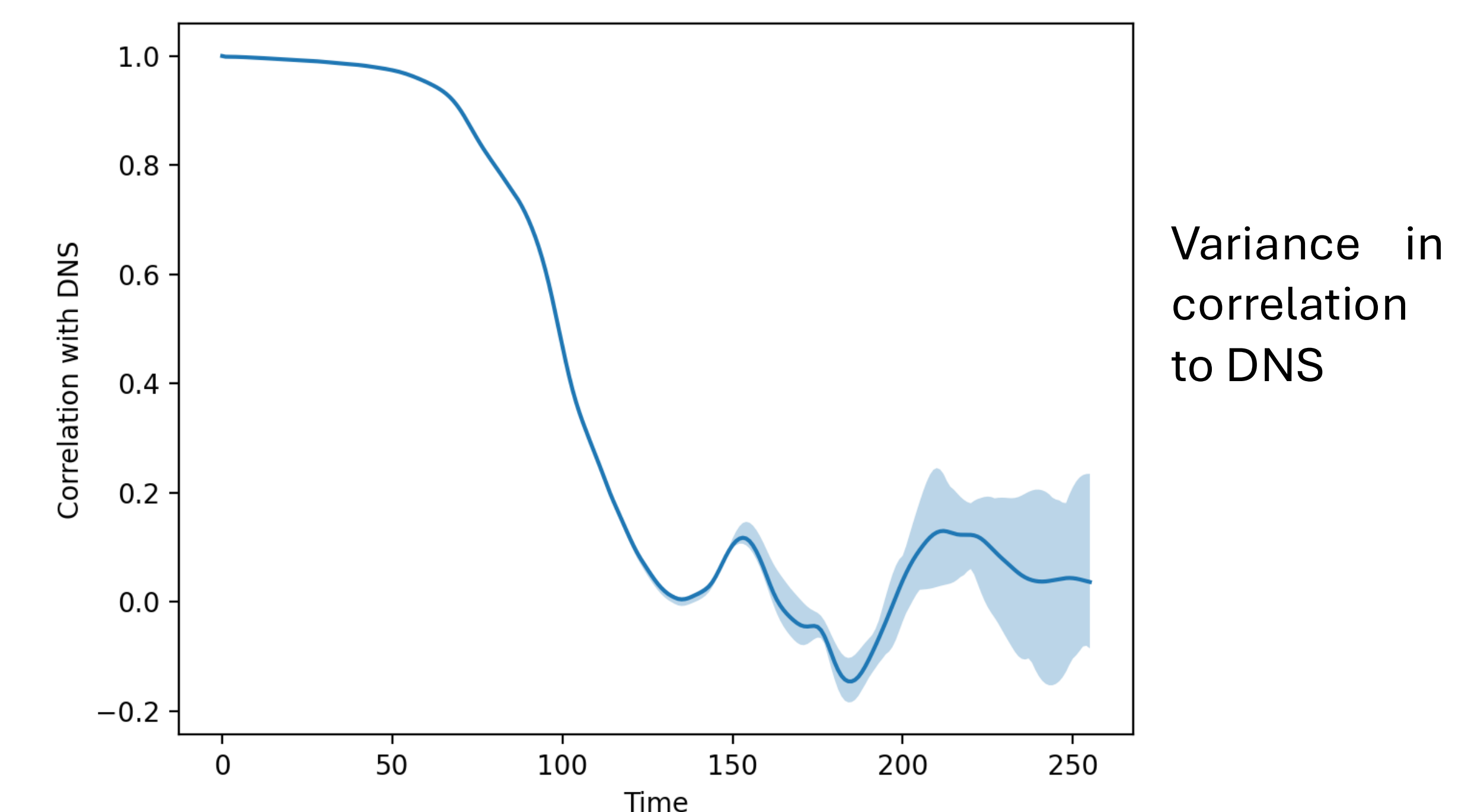
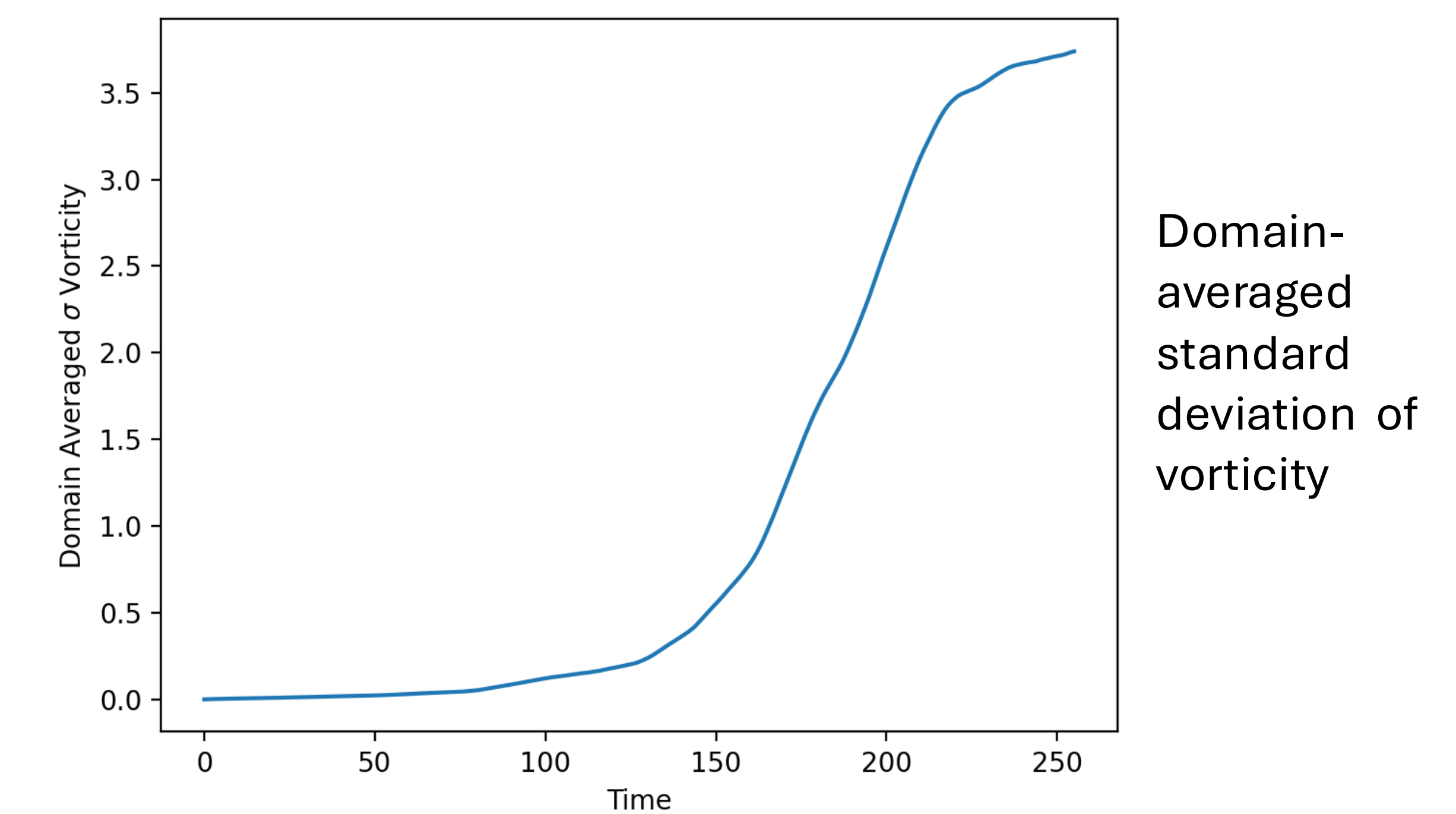
Turbulence KE production



The *a-priori* computed turbulence kinetic energy production PDF is visualized for the *a-priori* trained and *a-posteriori* trained ML models, as well as the true DNS and SMAG baseline.

Uncertainty Quantification

- Model is trained till convergence and then SGD iterates are used sample weights (SWAG technique)
- A-posteriori* model viability assessment is possible without having access to test data.



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